

Investigating Antibiotic Exposure and Risk of Severe Acute Graft Versus Host Disease in Patients Undergoing Allogeneic Hematopoietic Stem Cell Transplantation

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Several studies have shown that alteration of the microbiota, particularly in the gastrointestinal tract, can be associated with graft-versus-host disease (GvHD). Patients undergoing allogeneic hematopoietic stem cell transplantation (allo-SCT) usually get exposed to antibiotics (ATB), mainly in the peri-transplant period. Moreover, ATB, specifically those that target anaerobic bacteria, can alter the microbiota in a variety of body organs. This, in turn, renders several organs vulnerable to injury making them more prone to developing diseases such as GvHD. In this study, we evaluate whether the use of ATB, characterized by duration, timing and type, in the peri-transplant period, is associated with an increased incidence of acute GvHD (aGvHD) and aGvHD-related mortality.

In this retrospective study, we included 318 consecutive patients who underwent allo-SCT including haploidentical and cord blood transplantation between December 2012 and June 2018 at a single center. ATB exposure was collected and classified into 3 groups according to the date of initiation of ATB: from the start of conditioning to day - 1 of allo-SCT (early ATB exposure), from day 0 to neutrophil engraftment (late ATB exposure) and a third group of patients who did not receive ATB (no ATB exposure). ATB were only initiated if a patient develop fever or show signs of infection. Patients did not receive any prophylactic ATB. Exposure was further categorized as primary or adjunct. Primary exposure included the use of 4

classes of ATB: carbapenems, anti-pseudomonal penicillin, 4th and 5th generation cephalosporins and fluoroquinolones. Adjunct ATB comprised other ATB and was further divided into 2 groups according to anaerobic coverage. The median age at transplant was 55 years. The stem cell source was peripheral blood in 85%, bone marrow in 10% and cord blood in 5% of the patients. Ninety-nine percent of the patients received cyclosporine A as GvHD prophylaxis and 80% and 7% of the patients received (along with cyclosporine A), mycophenolate mofetil and methotrexate, respectively. In addition, 35% and 89% of the patients received post-transplant cyclophosphamide and anti-thymocyte globulin, respectively, as GvHD prophylaxis. The median time to neutrophil engraftment was 16 days post-transplant. The median follow-up was 85 months. 93.7% of the patients received ATB in the peri-transplant period with 64.5% of them in the early ATB exposure group, and 29.2% of them in the late ATB exposure group.

The 2-year overall survival and progression free survival were 74.3% and 63.6% in patients with early ATB exposure, compared to 79.5% and 70.8% in patients with late ATB exposure ($p=0.11$ and $p=0.07$ respectively). The 2-year cumulative incidence of non-relapse mortality was 16.5% in patients with early ATB exposure, compared to 15.1% in patients with late ATB exposure ($p=0.63$). The 180-days cumulative incidence of grade 2-4 and 3-4 aGvHD were 23.8% and 12.2% in patients with early ATB exposure, compared to 27.2% and 5.4% in patients with late ATB exposure ($p=0.64$ and $p=0.06$ respectively). In multivariate analysis, including the most important parameters associated with GvHD (stem cell source and donors, conditioning regimen, sex mismatch and patients age), early ATB initiation was the only parameter associated with a significantly higher risk of severe grade 3-4 aGvHD [HR 0.51 (0.28-0.90); $p=0.02$].

In conclusion, in the absence of any ATB prophylaxis, early initiation of ATB, before graft infusion, is associated with a significantly higher risk of severe grade 3-4 aGvHD. Weighing risk of morbidity and mortality associated with infections versus later on risk of developing aGvHD is essential. Hence, new strategies should be developed to risk stratify patients with fever and thus to avoid early non necessary ATB exposure especially in those who develop fever during anti-thymocyte globulin infusion. Studies evaluating such strategies will be necessary in the next years.

Disclosures

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Author notes

*Asterisk with author names denotes non-ASH members.

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